

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), m)$, where m denotes the Lebesgue measure on $[0, 1]$. Define the sequence of random variables X_n such that $X_1 \equiv 0$, $X_2 \equiv 13$ and $X_n(\omega) = n \mathbf{I}_{\{[0, \frac{1}{n}] \cup [1 - \frac{1}{n}, 1]\}}(\omega)$, $\omega \in \Omega$, $n \geq 3$. Let $X \equiv 0$.

- (i) Prove that $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.
- (ii) Prove: $\{X_n\}$ **not** uniformly integrable.
- (iii) Prove: X_n does **not** converge to X in L^1 as $n \rightarrow \infty$.

- (b) In general, under what conditions on the functions $\{X_n\}$ would
 “ $X_n \rightarrow X$ a.e. (P)” imply “ $X_n \rightarrow X$ in L^p ” for $p \geq 1$?

((3×10)+10=40 points)

- (2) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv \underline{3\mu_1 + 2\mu_2}$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$). Then, prove the following:

- (a) μ_3 is a measure.

- (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$. Also, express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

- (c) Let $\mu = \mu_3$ be defined as above where $\mu_1 = \text{Normal}(2, 3)$, $\mu_2 = \text{Poisson}(6)$. Find the Lebesgue decomposition of μ with respect to $\nu = m =$ the Lebesgue measure on $\mathbb{R}$, and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.

(10+(10+10)+15=45 points)

- (3) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$.

- (a) If $X > 1$ a.e. (P), then prove $E(\log X) \leq \log(EX)$.

- (b) Let Z be a *nonnegative* random variable with $0 < \text{Var}(Z) < \infty$ Prove:

$$P(Z > 0) \geq \frac{[E(Z)]^2}{E(Z^2)}.$$

[Use Cauchy-Schwarz Inequality]

(10+10 = 20points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$

Now want to find the R-N derivative $\frac{d\mu_a}{d\nu}$.

Consider that since

$$\begin{aligned}\mu_a(A) &= 3\mu_1(A) \\ &= 3 \int_A f_1 d\nu \\ &= \int_A 3f_1 dm\end{aligned}$$

where f_1 is the normal density and $3f_1$ is a nonnegative measurable function, the Radon-Nikodym theorem says that

$$3f_1 = \frac{d\mu_a}{dm} = \frac{d\mu_a}{d\nu}.$$

Hence,

$$\frac{d\mu_a}{d\nu} = \frac{3}{\sqrt{2\pi}(3)} \exp\left\{-\frac{(x-2)^2}{2(3)^2}\right\}. \quad \checkmark$$

③ X a v.v. on $(\Omega, \mathcal{F}, P)$.

(a) $X > 1$ a.e. (P) . Want to prove that

$$E[\log(X)] \leq \log(E[X]).$$

If f_1 is a non-negative measurable function, we have that

$$\mu_a \ll m. \quad \checkmark$$

Now let $B = (\mathbb{N} \cup \{0\})^c$ Then

$$\mu_s(B) = 2\mu_2(B)$$

$$= 2\mu_2((\mathbb{N} \cup \{0\})^c)$$

$$= 2(0)$$

$$= 0 \quad \text{since support of Poisson is } \{0, 1, 2, \dots\}$$

and

$$\nu(B^c) = m(\mathbb{N} \cup \{0\})$$

$$= 0$$

since m is the Lebesgue measure and B^c is a countable set. $\checkmark$

Hence, $\mu_a \ll \nu$ and $\mu_s \perp \nu$. Thus, the decomposition of μ into

$$\mu_a = 3\mu_1 \quad \text{and} \quad \mu_s = 2\mu_2$$

is the Lebesgue decomposition of μ wrt ν .

(c) $\mu = \mu_3$ and

$$\mu = 3\mu_1 + 2\mu_2$$

$$\mu_1 = \text{Normal}(2, 3)$$

$$\mu_2 = \text{Poisson}(6)$$

$$\nu = m$$

Consider that by definition + for $A \in \mathcal{F}$,

$$\mu_1(A) = \int_A f_1 dm \quad \text{and}$$

$$\mu_2(A) = \int_A f_2 d\mu_c \quad \text{where } \mu_c \text{ is the counting measure}$$

Here

$$f_1 = \frac{1}{\sqrt{2\pi}(3)} \exp \left\{ -\frac{(x-2)^2}{2(3)^2} \right\} = \frac{d\mu_1}{dm}$$

$$f_2 = \frac{e^{-6} 6^x}{x!} I_A(x) = \frac{d\mu_2}{d\mu_c}$$

Now, we want to decompose μ wrt ν .

Let

$$\begin{aligned} \mu_a &= 3\mu_1 \\ \mu_s &= 2\mu_2 \end{aligned}$$

$$\text{where } \mu_a + \mu_s = 3\mu_1 + 2\mu_2 = \mu \checkmark$$

Then, since we can write

$$3\mu_1(A) = \mu_a(A) = 3 \int_A f_1 dm = \int_A 3f_1 dm \quad \text{where } \forall A \in \mathcal{F}$$

Consider that $\mu_1 \ll \nu \Rightarrow \exists$ some h_1 a nonnegative measurable function s.t.

$$\mu_1(A) = \int_A h_1 d\nu \quad \forall A \in \mathcal{F}, \quad \text{with } h_1 = \frac{d\mu_1}{d\nu}$$

and $\mu_2 \ll \nu \Rightarrow \exists$ some h_2 a nonnegative measurable function s.t.

$$\mu_2(A) = \int_A h_2 d\nu \quad \forall A \in \mathcal{F}, \quad \text{with } h_2 = \frac{d\mu_2}{d\nu}$$

Then, by definition

$$\mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$$

$$= 3 \int_A h_1 d\nu + 2 \int_A h_2 d\nu$$

$$= \int_A 3h_1 d\nu + \int_A 2h_2 d\nu$$

$$= \int_A (3h_1 + 2h_2) d\nu \quad \text{by linearity of integrals}$$

$$= \int_A h_3 d\nu$$

where $h_3 = 3h_1 + 2h_2$ is a non-negative measurable function, so $h_3 = \frac{d\mu_3}{d\nu}$. Furthermore

$$h_3 = 3h_1 + 2h_2$$

$$= 3 \frac{d\mu_1}{d\nu} + 2 \frac{d\mu_2}{d\nu}$$

Thus,

$$\frac{d\mu_3}{d\nu} = 3 \frac{d\mu_1}{d\nu} + 2 \frac{d\mu_2}{d\nu}$$

a.e. (ν) (by the uniqueness of the R-N derivative.)

$$\begin{aligned}
\mu_3\left(\bigcup_{n=1}^{\infty} A_n\right) &= 3\mu_1\left(\bigcup_{n=1}^{\infty} A_n\right) + 2\mu_2\left(\bigcup_{n=1}^{\infty} A_n\right) \\
&= 3 \sum_{n=1}^{\infty} \mu_1(A_n) + 2 \sum_{n=1}^{\infty} \mu_2(A_n) \quad \text{since } \mu_1, \mu_2 \text{ measures} \\
&= \sum_{n=1}^{\infty} (3\mu_1(A_n) + 2\mu_2(A_n)) \\
&= \sum_{n=1}^{\infty} \mu_3(A_n) \quad \checkmark
\end{aligned}$$

Hence, μ_3 satisfies the conditions necessary to be a measure.

$$(b) \quad \mu_1 \ll \nu, \mu_2 \ll \nu \stackrel{?}{\Rightarrow} \mu_3 \ll \nu$$

$$\mu_1 \ll \nu \Rightarrow \nu(A) = 0 \Rightarrow \mu_1(A) = 0 \quad \forall A \in \mathcal{F}$$

$$\mu_2 \ll \nu \Rightarrow \nu(A) = 0 \Rightarrow \mu_2(A) = 0 \quad \forall A \in \mathcal{F}$$

Then if $\nu(A) = 0 \Rightarrow$

$$\mu_3(A) = 3\mu_1(A) + 2\mu_2(A) = 0 + 0 = 0$$

$$\Rightarrow \mu_3 \ll \nu. \quad \checkmark$$

Now want to express $\frac{d\mu_3}{d\nu}$ in terms of

$$\frac{d\mu_1}{d\nu} \quad \text{and} \quad \frac{d\mu_2}{d\nu} \dots$$

(b) What condition on $\{X_n\}$ would $X_n \rightarrow X$, a.e. (P)
 $\Rightarrow X_n \rightarrow X$ in LP.

• If $P(\Omega) < \infty$ and $\{X_n\}$ is UI.

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(2) $(\Omega, \mathcal{F})$ $\mu, \mu_1, \mu_2, \forall$ are σ -finite.

$$\mu_3 = 3\mu_1 + 2\mu_2$$

(a) WTS μ_3 is a measure...

$$\begin{aligned} (1) \mu_3(\emptyset) &= 3\mu_1(\emptyset) + 2\mu_2(\emptyset) \\ &= 3(0) + 2(0) \quad (\text{since } \mu_1, \mu_2 \text{ are measures}) \\ &= 0 \quad \checkmark \end{aligned}$$

$$(2) \mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$$

$$\left. \begin{array}{l} \text{where } 3\mu_1(A) \in [0, \infty) \\ 2\mu_2(A) \in [0, \infty) \end{array} \right\} \text{ since measures}$$

$$\Rightarrow \mu_3(A) \in [0, \infty) \quad \forall A \quad \checkmark$$

(3) Let $A_1, A_2, \dots \in \mathcal{F}$. Then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ since $\mathcal{F}$ a σ -algebra. Consider $\mu_3\left(\bigcup_{n=1}^{\infty} A_n\right) \rightarrow$

$$= \int n \mathbb{I} \left[\left[0, \frac{1}{n}\right] \cup \left[1 - \frac{1}{n}, 1\right] \right] dm$$

$$= n m \left(\left[0, \frac{1}{n}\right], \left[1 - \frac{1}{n}, 1\right] \right)$$

$$= n \left(\left(\frac{1}{n} - 0\right) + \left(1 - \left(1 - \frac{1}{n}\right)\right) \right)$$

$$= n \left(\frac{1}{n} + \frac{1}{n} \right)$$

$$= n \left(\frac{2}{n} \right)$$

$$= 2$$

This is true for all $T > 13$. Thus,

$$\sup_{n \geq 1} a_n(t) = 2 \quad \forall t.$$

Thus

$$\sup_{n \geq 1} a_n(t) \rightarrow 2 \quad \text{as } t \rightarrow \infty$$

$$\not\rightarrow 0 \Rightarrow \text{not UI.}$$

(iii) Prove $\{X_n\}$ does not converge in L^1 as $n \rightarrow \infty$.

Consider for $n > 2$

$$\int |X_n|' dm = \int n \mathbb{I} \left[\left[0, \frac{1}{n}\right] \cup \left[1 - \frac{1}{n}, 1\right] \right] dm$$

$$= n m \left(\left[0, \frac{1}{n}\right] \cup \left[1 - \frac{1}{n}, 1\right] \right)$$

$$= n \left(\frac{2}{n} \right) = 2$$

$$\text{Thus, } \lim_{n \rightarrow \infty} \int |X_n|' dm = 2$$

Hence, $X_n \rightarrow X$ in L^1 . $\neq 0$. ✓

same work as before

① $([0,1], \mathcal{B}([0,1]), m)$ $m = \text{Lebesgue}$

X_n s.t. $X_1 \equiv 0$, $X_2 \equiv 13$, and $X_n(\omega) = n \mathbb{I}_{\{[0, \frac{1}{n}] \cup [1 - \frac{1}{n}, 1]\}}(\omega)$
when $n \geq 3$

$$X \equiv 0$$

(i) Prove $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.

When $\omega = 0$

$$X_n(\omega) \rightarrow \infty \text{ as } n \rightarrow \infty$$

when $\omega = 1$

$$X_n(\omega) \rightarrow \infty \text{ as } n \rightarrow \infty$$

when $\omega \in (0,1)$

$$X_n(\omega) \rightarrow 0 \text{ as } n \rightarrow \infty \quad (\text{since for all } \omega,$$

$\exists$ some n_0 s.t. when $n > n_0$ $\frac{1}{n_0} < \omega < 1 - \frac{1}{n_0}$).

Thus,

$$X_n(\omega) \rightarrow 0 \text{ as } n \rightarrow \infty \quad \forall \omega \in (0,1).$$

Let $B = (\{0\}, \{1\})$. Then $\mu(B) = m(\{0\}, \{1\}) = 0$

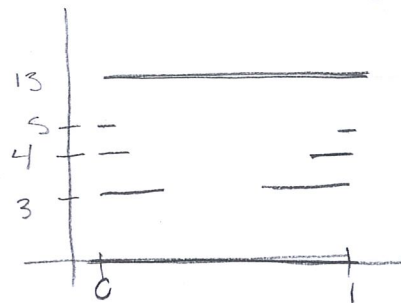
(since Lebesgue measure). Hence. $X_n \rightarrow X$ a.e. (m).

(ii) Prove $\{X_n\}$ not uniformly integrable... Consider

$$a_n(T) = \int |X_n| \cdot \mathbb{I}_{\{|X_n| > T\}} dm \quad \text{for some } T > 0$$

when $n > T$ (and $n > 2$)

$$a_n(T) = \int n \mathbb{I}_{[0, \frac{1}{n}] \cup [1 - \frac{1}{n}, 1]} dm \longrightarrow$$



Consider that $\phi(x) = -\log(x)$ with $x > 0$ is a convex function. Then by applying Jensen's inequality (w/ $\int |x| dP < \infty$), we get

$$\phi(E[X]) \leq E[\phi(X)]$$

$$\Rightarrow -\log(E[X]) \leq E[-\log(X)]$$

$$\Rightarrow \log(E[X]) \geq -E[-\log(X)]$$

$$\Rightarrow E[\log(X)] \leq \log(E[X]). \quad \checkmark$$

This will also be true in the case when $\int |x| dP = \infty$ since

$$E[X] = \infty \Rightarrow \log(E[X]) = \infty$$

$$\Rightarrow E[\log(X)] \leq \log(E[X]) = \infty. \quad \checkmark$$

b) Now want to show that for Z a non-negative random variable w/ $0 < \text{Var}[Z] < \infty$,

$$\int z dP = P(Z > 0) \geq \frac{E[Z]^2}{E[Z^2]} = \frac{(\int z dP)^2}{(\int z^2 dP)}$$

suggests using Cauchy-Schwarz

$$\int |fg| d\mu \leq \left(\int |f|^2 d\mu \right)^{\frac{1}{2}} \left(\int |g|^2 d\mu \right)^{\frac{1}{2}}$$

Let $f = z^{\frac{1}{2}}$ $g = z$ and

~~$\int z^{\frac{1}{2}} z$~~

$$\left(\int z dP\right) \left(\int z^2 dP\right) = \left(\int z dP\right)^2$$

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$$\frac{94}{105} \rightarrow \left(\frac{94}{100}\right)$$